

Daniel W Liu

Computer Science and Finance Student

[Personal Portfolio](#) | D347Liu@uwaterloo.ca | [LinkedIn/DanielWLIu](#) | [GitHub/DanielWLIu07](#) | +1 (647) 679-0336

EDUCATION

University of Waterloo | *Double Major in Computer Science and Finance*
Bachelor of Computing and Financial Management (Co-op) | **GPA: 3.94/4.0**

Sept 2025 – April 2029
Waterloo, ON

EXPERIENCE

Software Developer Intern

May 2026 – Aug 2026

PlayStation Sony Interactive Entertainment

Waterloo, ON

- Engineered a **34-endpoint** Express.js admin API layer authorization proxy, surfaced by a React portal enabling admins to manage roles, permissions, and access tokens for **2,500+ developers** through GitHub-backed RBAC.
- Developed an authorization microservice using YAML inheritance graphs to derive roles from explicit GitHub team bindings stored in a **13-table** PostgreSQL database, propagating permission changes to **12+ developer tools**.
- Migrated publishing for **16** Kubernetes Helm charts from Jenkins to GitHub Actions, providing AWS ECR access through CloudFormation-managed IAM roles and cutting publish time by **56%** with Docker build optimizations.

Software Engineering Intern

Jan 2026 – Apr 2026

Wedge YC S25

San Francisco, CA

- Shipped a full-stack autonomous voice agent platform using Next.js, Postgres, and LiveKit that calls health insurers to check claim status, iterating weekly with **3 hospital clients** to deploy the platform across **50+ clinical sites**.
- Designed a recursive IVR-mapping engine that re-dials insurers to explore unwalked phone menu branches, merging every call into one canonical interactive React call map spanning a **4,839-number** directory of **2,841 payers**.

Software Engineering Intern

Sept 2025 – Dec 2025

HOSA Canada

Toronto, ON

- Engineered a scalable full-stack resource sharing platform with Next.js, AWS S3, and PostgreSQL that streams written test events through Apache Kafka to serve **9,000+ Canadian students**, verified by **1,098 Jest tests**.
- Deployed a rate-limited MERN workshop sign-up service processing **19,943 registrations** across **347 workshop sessions**, draining Redis-backed job queues with PM2 cron workers to automate SendGrid email notifications.

Machine Learning Research Intern

Jul 2025 – Aug 2025

Pearl Lab Holland Bloorview Kids Rehabilitation Hospital

Toronto, ON

- Developed real-time computer vision for *Bootle Blast*, a gamified rehab platform clinically tested at **3+ hospitals**, fusing depth cameras with OpenCV and PyTorch CNN inference to reconstruct children's motor movement in 3D.
- Trained ML YOLOv8 detection models on a custom dataset built from **250+ environments**, benchmarked using NumPy and pandas to beat classical segmentation by **46.8% mean average precision** at **sub-10 ms** inference.

PROJECTS

rainbolt.ai | *Next.js, Three.js, FastAPI, OpenAI CLIP, Pinecone, LangChain, Google Cloud*

Devpost GitHub

- Awarded **Best UI Hack** at UofT Hack the Valley X for building an AI geolocation teaching platform with Next.js, Three.js, Zustand, and Gemini, delivering interactive 3D globe visualizations and narrated geographic reasoning.
- Architected a RAG pipeline that embeds street-level imagery with OpenAI CLIP into a Pinecone index of **955,199+ images**, retrieving matches through a FastAPI backend to ground every guess in visual evidence.

Pomme | *Arduino UNO Q, C++, React.js, Edge Impulse, SLAM, Socket.IO, MongoDB*

Devpost GitHub

- Won **Qualcomm's Build at the Edge award** at Hack the 6ix with an autonomous fruit-picking robot pairing a Linux AI chip and a real-time MCU to run navigation, computer vision, and motor control in a **5-watt envelope**.
- Trained a YOLOv8 ripeness detector on procedurally generated synthetic data, quantized it **4x smaller** to **INT8** for on-device inference, and surfaced its detections live in a React.js/Three.js dashboard with LiDAR SLAM maps.

Multithreaded Voxel Engine | *C++20, OpenGL 4.1, GLFW, GLM, Dear ImGui, FastNoiseLite, CMake* *GitHub*

- Built a C++20/OpenGL engine streaming a **40-million-voxel** world at **229 FPS** in 188 MB of RAM, using greedy meshing to cut triangles **18x** and shrink GPU memory from 48 to 12.5 MB with a CI gate blocking regressions.
- Designed a 9-thread chunk pipeline meshing terrain at **2,200 chunks/sec** with **8.4x parallel efficiency** and occlusion culling drawing up to **70x** less hidden geometry, verified with thread and address sanitizers in CI.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, JavaScript, TypeScript, Java, SQL, Bash, HTML, CSS

Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, OpenGL, Three.js, Tailwind CSS

Cloud/Data: AWS, PyTorch, OpenCV, PostgreSQL, MongoDB, Redis, Supabase, Drizzle, pandas, NumPy

DevOps/Tools: Kubernetes, Docker, Jenkins, GitHub Actions, pytest, unittest, Git, Linux, CI/CD, Jest, Agile, Jira